Research Design and Methodology

Comparative Study of Explainable AI Models for Pneumonia Detection from Chest X-rays

Pavani Kumari Yamala

FA26 - COMP-699-A - PROFESSIONAL SEMINAR

Professor David Pitts

October 18, 2026

1. Research Topic and Research Problem

I plan to study how two deep learning models classify chest X-ray images as pneumonia or normal and how their Grad-CAM explanations differ. My main interest is not only whether a model makes a correct prediction but also which areas of an image appear important to that prediction. I want to compare the models under the same experimental conditions and examine cases where they make different predictions or produce different visual explanations.

Deep learning has been used to classify chest X-rays, including research on pneumonia detection by Rajpurkar et al. (2017). However, strong classification performance does not always mean that a model has learned the right visual patterns. Zech et al. (2018) found that pneumonia detection models could use information related to the hospital where an image was collected, which raises concerns about whether a model is learning disease-related features or other patterns. I want to investigate this issue by comparing prediction results and visual explanations in a controlled experiment. The study will help me understand both the useful features and the limitations of these models.

2. Research Question

My primary research question is:

How do ResNet50 and DenseNet121 compare in pneumonia detection performance, and how do their Grad-CAM explanations differ on the same chest X-ray images?

I will use two secondary questions to guide the analysis:

1. How often do the models agree or disagree on their predictions, and how do their confidence scores compare in these cases?
2. When both models process the same X-ray, do their Grad-CAM maps highlight similar or different image regions?

These questions are narrow enough for a semester project and can be investigated using Python, a public dataset, pretrained models, and measurable outputs.

3. Working Hypothesis or Tentative Answer

My working hypothesis is that ResNet50 and DenseNet121 will show differences in classification performance and in the image regions highlighted by Grad-CAM. I expect the models to agree on many images but to disagree on some difficult cases. I also expect that some correct predictions may have low confidence or visual explanations that are difficult to interpret.

I do not assume that either model will perform better before testing them. I will use the same dataset, image preprocessing, test images, and evaluation metrics for both models. The results will determine whether my expectations are supported. If the models perform similarly, or if their visual explanations show no clear pattern, I will report those findings rather than forcing a difference.

4. Significance of the Research Question

I believe this research matters because a model's accuracy alone cannot explain how it reaches a prediction. In medical image analysis, a model might classify an X-ray correctly while relying on an irrelevant image feature. Zech et al. (2018) showed why differences

between hospitals and image sources can affect model performance. This makes it useful to examine both classification results and visual explanations.

My project will not establish whether either model is safe for clinical use. Instead, it will provide a small, controlled comparison that shows how performance metrics, confidence scores, and Grad-CAM maps can give different kinds of information about model behavior. I hope this study will help readers understand why explainability should be examined alongside predictive performance rather than treated as proof that a model is medically reliable.

5. Review of Relevant Sources

I will use previous research to guide my model selection, evaluation methods, and interpretation of results. Kermany et al. (2018) demonstrated the use of deep learning for medical image classification and included a pediatric chest X-ray dataset for pneumonia detection. Their work provides a useful starting point for selecting a public dataset and understanding how image classification can support medical research. Rajpurkar et al. (2017) also demonstrated pneumonia detection using a deep convolutional neural network called CheXNet. These studies show the potential of deep learning, but they do not remove the need to evaluate models carefully on the data used in a new experiment.

I will use Grad-CAM because it generates a heatmap showing image regions that contribute to a model's prediction. Selvaraju et al. (2017) introduced Grad-CAM as a method for producing visual explanations from deep neural networks. However, I will not treat a visually convincing heatmap as proof that a model has learned medically meaningful features. Adebayo et al. (2018) showed that some saliency maps can appear reasonable even when they do not properly reflect a model's learned behavior. This finding supports my decision to examine Grad-CAM maps together with prediction results and model disagreements.

I will also consider research on model reliability and dataset bias. Zech et al. (2018) found that pneumonia detection performance varied across hospital systems, showing that models may use patterns related to image sources rather than disease alone. Roberts et al. (2021) reviewed common problems in machine learning studies involving chest imaging and highlighted concerns about biased datasets, weak evaluation methods, and limited reproducibility. Together, these studies show why I need to document preprocessing, use a consistent test set, and avoid making claims beyond the evidence.

The research I reviewed provides methods for classification and visual explanation, but my project will focus on a specific comparison: two models evaluated on the same images, followed by a closer examination of their disagreements and Grad-CAM patterns. My contribution will be a small, reproducible experiment that brings these measurements together. I will describe the limits of this comparison and will not claim that visual similarity alone proves that a model is trustworthy.

6. Evidence

I will use a public pediatric chest X-ray dataset described by Kermany et al. (2018). The dataset contains images labeled as normal or pneumonia and has been used in published research. I will confirm the available files and labels before beginning the full experiment. The dataset is the source of the evidence; the predictions, scores, metrics, and explanation maps produced from it will be the evidence I analyze.

First, I will collect classification evidence from ResNet50 and DenseNet121. This will include each model's predicted class, confidence score, and whether the prediction matches

the dataset label. I will calculate accuracy, precision, recall, F1-score, and ROC-AUC. These measurements will help me compare overall performance and identify differences in the models' ability to detect pneumonia.

Second, I will collect evidence about model agreement. I will record the number and percentage of test images for which both models predict the same class and the number for which their predictions differ. I will review selected disagreement cases with their true labels and confidence scores. This will help me identify cases that deserve closer examination.

Third, I will generate Grad-CAM maps for selected images and compare them side by side. I will record whether the highlighted regions appear similar or different and whether the heatmaps extend beyond the visible lung areas. These observations will be descriptive, not medical judgments. Grad-CAM provides an approximate visual explanation, not proof that a model is focusing on the correct medical evidence (Selvaraju et al., 2017).

The main limitation is that the dataset's labels and image sources may contain errors or bias. The images also represent a specific population and collection setting. Therefore, I will limit my conclusions to the selected dataset and will not assume that the results apply to all patients or hospitals.

7. Computational Method for Producing Evidence

I will use Python and PyTorch to build the experimental workflow. I plan to use pretrained ResNet50 and DenseNet121 models and adapt their final classification layers for two classes: normal and pneumonia. I will use the same training and evaluation process for both models so that the comparison is fair.

I will first download the dataset from its public source and inspect the folder structure, image formats, and class labels. I will check for unreadable images and duplicate files where possible. I will resize images to the input size required by the selected models, convert them to tensors, and apply the normalization required by their pretrained weights. I will keep the training, validation, and test sets separate and will not use test results to select model settings.

Next, I will train or fine-tune the classification layers using the training data and use the validation data to select settings. I will fix random seeds where possible and record the model settings, software versions, and training details. After the model settings are finalized, I will evaluate both models on the same held-out test images.

I will use scikit-learn to calculate the evaluation metrics and Matplotlib to prepare charts and visual comparisons. I will implement Grad-CAM for the final convolutional layer of each model and generate heatmaps for selected predictions. I will also create a comparison script that identifies agreement and disagreement cases and saves their predictions, confidence scores, labels, and explanation images.

The software I develop will include scripts for data loading, preprocessing, model evaluation, Grad-CAM generation, and result reporting. I may add a small Streamlit interface to view selected images and compare the models, but the main deliverable will be the experimental code and analysis. This workflow connects the research question to the data, computational methods, and evidence needed to evaluate my hypothesis.

8. Preliminary Evidence and Feasibility Demonstration

My first pilot experiment will test whether I can load a small group of chest X-ray images, apply the required preprocessing, and pass the images through both pretrained models. I will then generate one Grad-CAM map from each model and save the outputs for comparison.

This will help me confirm that the main parts of the proposed workflow can run before I begin the full experiment.

At the time of preparing this research design, I have not established measured results from running this pilot. I will therefore not report invented accuracy scores, predictions, or heatmaps. The first feasibility check will record the number of images loaded, the class labels read successfully, whether both models produce outputs, and whether the Grad-CAM images are saved. I will include the actual output and a sample visualization in the final document after running the code.

A successful pilot would demonstrate that the data-loading, prediction, and visualization steps work on a small sample. It would not demonstrate that either model is accurate, that Grad-CAM highlights medically correct regions, or that the full research hypothesis is supported. Those questions require the planned evaluation on the held-out test set.

9. Anticipated Claims, Reasons, and Evidence

My tentative main claim is that comparing model performance with visual explanations provides a more complete view of pneumonia classification than accuracy alone. I will test this claim using three types of evidence.

The first supporting reason may be that ResNet50 and DenseNet121 have different strengths in classification. I will use accuracy, precision, recall, F1-score, and ROC-AUC from the same test set to determine whether there is a measurable difference. If their results are similar, I will report that instead.

The second supporting reason may be that the models make different predictions on some images. I will use a table of agreement and disagreement cases, including the true labels and confidence scores, to determine how often this occurs. The third supporting reason may be that the models produce different Grad-CAM patterns for the same image. I will compare selected heatmaps and record visible similarities and differences. These are tentative claims, and I will accept, revise, or reject them based on the evidence.

10. Qualifications, Limitations, and Alternative Explanations

I recognize several limits that could affect my findings. First, the selected dataset represents a specific population and source, so the results may not generalize to adults or other hospitals. Second, the dataset labels may not capture every clinical detail, and image quality may vary. I will report these limitations and avoid presenting model predictions as medical diagnoses.

Third, a model may learn image markings, borders, or other source-related patterns instead of features associated with pneumonia. Zech et al. (2018) and Roberts et al. (2021) explain why dataset bias and weak external validation can produce misleading results. I will examine Grad-CAM maps for signs of attention outside the lung areas, but I will not assume that a heatmap proves the reason for a prediction. Adebayo et al. (2018) also showed that visual explanations need careful evaluation.

I will use the same test images and preprocessing rules for both models, document the model settings, and keep the test set separate from training and validation. I will save the code and instructions needed to repeat the experiment. Since the project uses a public, de-identified dataset, I do not plan to collect personal health information. I will still follow the dataset's terms of use and explain that the study is for research and education, not clinical use.

11. Research Schedule

I will complete the project in stages, with December 8, 2026, as my final deadline.

- October 19–25: Confirm the dataset, review the research sources, and create the GitHub repository.
- October 26–November 1: Implement data loading, preprocessing, and the initial model setup.
- November 2–8: Run the pilot experiment and save its actual output and sample Grad-CAM maps.
- November 9–17: Complete the model evaluation using the same held-out test images.
- November 18–24: Compare metrics, confidence scores, disagreement cases, and Grad-CAM patterns.
- November 25–30: Interpret the evidence, review limitations, and develop the main argument.
- December 1–4: Complete the research paper and organize the code, README, dependencies, and reproducibility instructions.
- December 5–7: Revise the paper, check citations, and verify the repository instructions.
- December 8: Submit the final paper and supporting project materials.

This schedule gives me time to test the method before completing the full analysis. I will use the results to guide my final claims rather than assuming that my initial hypothesis must be correct.

References

Adebayo, J., Gilmer, J., Muelly, M., Goodfellow, I., Hardt, M., & Kim, B. (2018). Sanity checks for saliency maps. Advances in Neural Information Processing Systems, 31, 9505–9515. https://arxiv.org/abs/1810.03292

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (pp. 770–778). https://doi.org/10.1109/CVPR.2016.90

Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (pp. 4700–4708). https://doi.org/10.1109/CVPR.2017.243

Kermany, D. S., Goldbaum, M., Cai, W., Valentim, C. C. S., Liang, H., Baxter, S. L., McKeown, A., Yang, G., Wu, X., Yan, F., Dong, J., Prasadha, M. K., Pei, J., Ting, M. Y. L., Zhu, J., Li, C., Hewett, S., Dong, J., Ziyar, I., ... Zhang, K. (2018). Identifying medical diagnoses and treatable diseases by image-based deep learning. Cell, 172(5), 1122–1131.e9. https://doi.org/10.1016/j.cell.2018.02.010

Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C., Shpanskaya, K., Lungren, M. P., & Ng, A. Y. (2017). CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. arXiv. https://arxiv.org/abs/1711.05225

Roberts, M., Driggs, D., Thorpe, M., Gilbey, J., Yeung, M., Ursprung, S., Aviles-Rivero, A. I., Etmann, C., McCague, C., Beer, L., Weir-McCall, J. R., Teng, Z., Gkrania-Klotsas, E., Rudd, J. H. F., Sala, E., & Schönlieb, C.-B. (2021). Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. Nature Machine Intelligence, 3, 199–217. https://doi.org/10.1038/s42256-021-00307-0

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In Proceedings of the IEEE International Conference on Computer Vision (pp. 618–626). https://doi.org/10.1109/ICCV.2017.74

Zech, J. R., Badgeley, M. A., Liu, M., Costa, A. B., Titano, J. J., & Oermann, E. K. (2018). Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. PLOS Medicine, 15(11), Article e1002683. https://doi.org/10.1371/journal.pmed.1002683